

ABDELRAHMAN MOHAMMED

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EDUCATION

German University In Cairo

Bachelor of Engineering in Computer Science and Engineering
Bachelor Thesis: "Logic and Ontology of Ability" - Grade: **A+**

Cairo, Egypt
October 2021 - (July 2026)

EXPERIENCE

TheLearningMate

Junior Software Engineer

Remote
August 2025 - Present

- Increased platform usage by 30% by owning a **Google Classroom** integration end-to-end, designing an event-driven pipeline with **AWS Lambda**, **Google Pub/Sub**, and **Inngest** workflows from system design doc through production
- Built a serverless PDF generation service on **AWS Lambda** using **Puppeteer/Chromium** to render LaTeX and HTML exam content, with **S3** presigned URLs and timing-safe token validation for secure delivery
- Developed full-stack features using **Next.js**, **TypeScript**, and **PostgreSQL** with **Drizzle**, enhancing user experience and application performance

Astronomic

Junior Software Engineer

Remote
May 2024 - January 2025

- Improved data quality to below 1% invalid records across 10M+ investor and company records by implementing similarity algorithms for de-duplication and error correction
- Reduced pipeline runtime by 90% by optimizing data processing with **Node.js**, **TypeScript**, and advanced **MongoDB** aggregations

Learning Management System

Project Manager & Backend Software Engineer

Freelance Client Project
July 2024 - May 2025

- Architected and implemented a scalable backend using **Express.js** and **PostgreSQL**, handling 1,000+ daily active users with 99.9% uptime
- Led technical planning and database design, implementing role-based access control for students, teachers, and administrators
- Owned the product lifecycle end-to-end, translating client requirements into technical specifications, prioritizing features, and iterating based on direct user feedback

German University in Cairo

Junior Teaching Assistant

Cairo, Egypt
February 2023 - May 2024

- Guided 40+ students across **OOP** in **Java**, logic programming in **Prolog**, and functional programming in **Haskell** during weekly lab sessions

TECHNICAL SKILLS

Programming Languages:	Python, Typescript, Javascript, Java, C++, C
Technologies:	Next.js, React.js, Node.js, Express.js, Django, Flask, AWS Lambda, Docker, Git
Cloud & Infra:	Google Cloud Pub/Sub, Inngest, Puppeteer, S3
Databases:	PostgreSQL, MongoDB, MySQL, Drizzle, Prisma, Mongoose

PROJECTS

IGCSE CS PseudoCode Web Compiler

<https://www.pseudocode-compiler.sherlemious.com>

Built a browser-based pseudocode interpreter for the Cambridge IGCSE syllabus using a custom **ANTLR4** grammar and tree-walking interpreter in **TypeScript**; fully client-side, executing the parse tree directly.

Mern Stack Tourism Website

github.com/Advanced-computer-lab-2024/Are-we-there-yet

Managed team of 10 developers to build a **MERN** stack tourism platform with Stripe integration and real-time booking, achieving 100% score and performance bonus.

Database Engine

github.com/Sherlemious-GUC-Projects/DB2-Project1

Built a custom database engine in **Java** featuring B+ Tree indexing, efficient data pagination, and ACID-compliant transactions. Implemented query optimization, achieving $O(\log n)$ search complexity.

LEADERSHIP AND AWARDS

ECPC Qualifications Participant

2022, 2023, 2024, 2025

The Intelligent Question (TIQ) - President

September 2023 - July 2024

Directed a debate club, organised events, and mentored members to improve critical thinking and public speaking skills.